



P35

**P35 Production-Grade
Multi-Agent Engineering
Architecture: A Five-Layer Stack,
a Six-Form Morphology
Spectrum, a Harnessed
Production SOP, and a Ten-Class
Component Field Specification**



P35 Production-Grade Multi-Agent Engineering Architecture: A Five-Layer Stack, a Six-Form Morphology Spectrum, a Harnessed Production SOP, and a Ten-Class Component Field Specification

Supplemental volume P35 • consolidating the multi-agent ecosystem, which by 2025–2026 had differentiated into a five-layer industrial stack, into one constructible engineering blueprint • UDOS Reasoning Engine v7.7.1 • Fang Wenxin • 2026-09-21

First said clearly in one sentence: today's "multi-agent" is long no longer one model, but a vertically integrated industry chain from **model substrate** → **protocols** → **development/enterprise platforms** → **products** → **C-end forms**; this paper uses the **five-layer stack** to position what each layer sells, uses the **six-form** × **nine-dimension matrix** to distinguish six Agent forms, uses the **production SOP of 8 stages** × **5**

human-review gates + 1 red-team gate + 1 version gate to write "how to build a launchable Agent" as a pipeline, then uses the **ten-class component field specification tables** to turn each layer's abstraction into configurable fields. **The full text repeatedly declares: all new architectures and fields are design proposals • to be implemented; external numbers uniformly marked per ledger three states (existence verified / paper • company self-report not independently reviewed / no evidence found), not endorsing vendors, not fabricating benchmarks.**

Evidence caliber (declared only once in full text): this paper is an industry architecture **specification and positioning** document, **containing no new UDOS measured numbers**. The existence of external industry facts and paper claims was item-by-item verified on 2026-09-21 per EVIDENCE_LEDGER_v771.md; among them MCP 2026-07-28 stateless core revision, A2A v1.0 Signed Agent Card, AAIF establishment and A2A merge timeline, Volcano Engine product matrix, Microsoft/OpenAI dual SDK Handoff standardization, marked  [existence verified]; Warp-Cortex concurrency/VRAM/capacity, Anthropic R&D automation share, IDC market share, GPT-5.3-Codex self-reference statements, marked  [paper/company disclosure • not independently reviewed] or [via media retelling • original not pulled]; any unverified capability numbers uniformly marked [CITATION NEEDED]. All layering, matrices, SOP gates, field specifications proposed by this paper are **design proposals • to be implemented**, not written as landed measurements. Verification date 2026-09-21.

Abstract

The 2026 multi-agent industry has already left the single-dimensional stage of "taking one large model to chat," differentiated into a **five-layer vertically integrated stack**: L1 model substrate, L2 protocol layer (MCP / A2A / SAEP), L3 development and enterprise platforms, L4 product layer, L5 C-end forms. This paper's task is not to repeat once more "Agents are cool," but to **engineer-wise take apart** this stack: ① using one "five-layer responsibilities × key products × inter-layer interfaces" table to pin the industry chain on the chart; ② using the nested relationship "sub-agent $\subset$ Agent $\subset$ super Agent legion" and the L0–L5 autonomy ladder (modeled on SAE J3016 / Epoch AI AL0–AL5) to place coordinates for each form; ③ using one "six forms × nine dimensions" matrix to list at once the genealogy from Chatbot to super Agent legion, and point out three key leaps (SOP→goal, A2A catalysis, memory architecture catalysis); ④ writing the process of building a launchable Agent as an 8-stage × 4-role swimlane SOP, with 5 human-review gates + 1 red-team gate + 1 version gate, emphasizing stage 5 acceptance "mechanically decidable," not relying on subjective scoring; ⑤ consolidating the whole abstraction into ten-class component field specification tables (Agent / MCP tools / A2A communication / TransferBundle / circuit breaker / BFT-lite committee / internal market settlement / governance audit / evidence grading / model selection). The full text's all numbers marked per ledger three states, all new architectures marked "design proposal • to be implemented," and at the end gives dross, failure boundaries and falsifiability conditions.

Keywords: multi-agent engineering; five-layer stack; MCP/A2A/SAEP; autonomy ladder L0–L5; form genealogy; production SOP; human-review gates; component field specifications; design proposals; evidence three states

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** in the market the word "Agent" is overused for everything from chatbots to million-Agent legions, causing selection to conflate "can it be autonomous," "does it need human

review," "who manages it"; lacking one construction drawing that layers the industry chain, spectra the forms, stages the production process.

- **Core argument:** multi-agent is an engineering stack **vertically integrating five layers**, different forms only different slices of this stack on the three axes "autonomy × collaboration × governance"; the key to building production-grade Agents is not smarter models, but the three engineering things **protocol standardization + gated production + componentized configuration**.
- **Evidence threads:** protocol side anchored on MCP 2026-07-28 stateless core revision, A2A v1.0 Signed Agent Card, AAIL unified governance (✅ verified); industry side corroborated by Volcano Engine four-layer product matrix, dual SDK standardization of Handoff/Agent-as-Tool (✅ verified); frontier numbers cited downgraded per three states for Warp-Cortex (single-author preprint self-report), Anthropic R&D automation index (company disclosure), IDC share (media retelling).
- **Contribution:** one five-layer stack interface table; one L0–L5 autonomy ladder and concept nesting chart; one six-form × nine-dimension matrix; one 8-stage SOP and gate entry/exit table; ten-class component field specification tables; and a set of falsifiable failure boundaries.

1 Why This Industry Drawing Is Needed

In 2024 saying "Agent," default meant "one large model chat box that can call tools"; by 2026, the word has been stretched into a continuous spectrum from **C-end small assistants** to **590-Agent national-level policy simulation** (MegaAgent, ACL 2025 Findings, ✅ existence verified). The cost of word overuse is engineering decision distortion: some deploy full zero-trust RBAC on a "read-only Q&A Chatbot," others let a "goal-driven autonomous Agent" run in a bare container with no fault recovery strategy.

This paper's position: **multi-agent is not one product, but an industry chain**. To explain it clearly needs four things—layering (what each layer sells, how to connect), spectra (where six forms differ), flow (how many steps in the pipeline to build one, where to set gates), fragment (consolidating abstraction into configurable fields). The following five sections correspond to five hard deliverables, delivered item by item.

Chapter summary: production-grade multi-agent's first-principle problem is not "how strong the model," but "how this stack layers, how forms spectrum, how production gates, how components configure." The following five sections are the answers to these four questions.

2 Deliverable One: Five-Layer Vertically Integrated Layered Architecture

First the full picture.

distributed along the **zero code → low code → high code → fully managed** gradient (the above only describes public product positioning, not endorsing vendors).

- **L4 product layer:** consolidates platform-built Agents into three role types—work Agent (docs/code/data analysis), companion Agent (schedules/social/emotional), phone assistant Agent (notifications/device control/multi-device collaboration).
- **L5 C-end forms:** the entries users actually contact—phone assistants (App/mini program), work platforms (PC/Web), AI companions (IM/social), browsers (plugins), voice entries (smart speakers). Entries spread along four modalities **touch/voice/vision/natural language**, interaction evolves along three tiers **conversation → delegation → autonomy**.

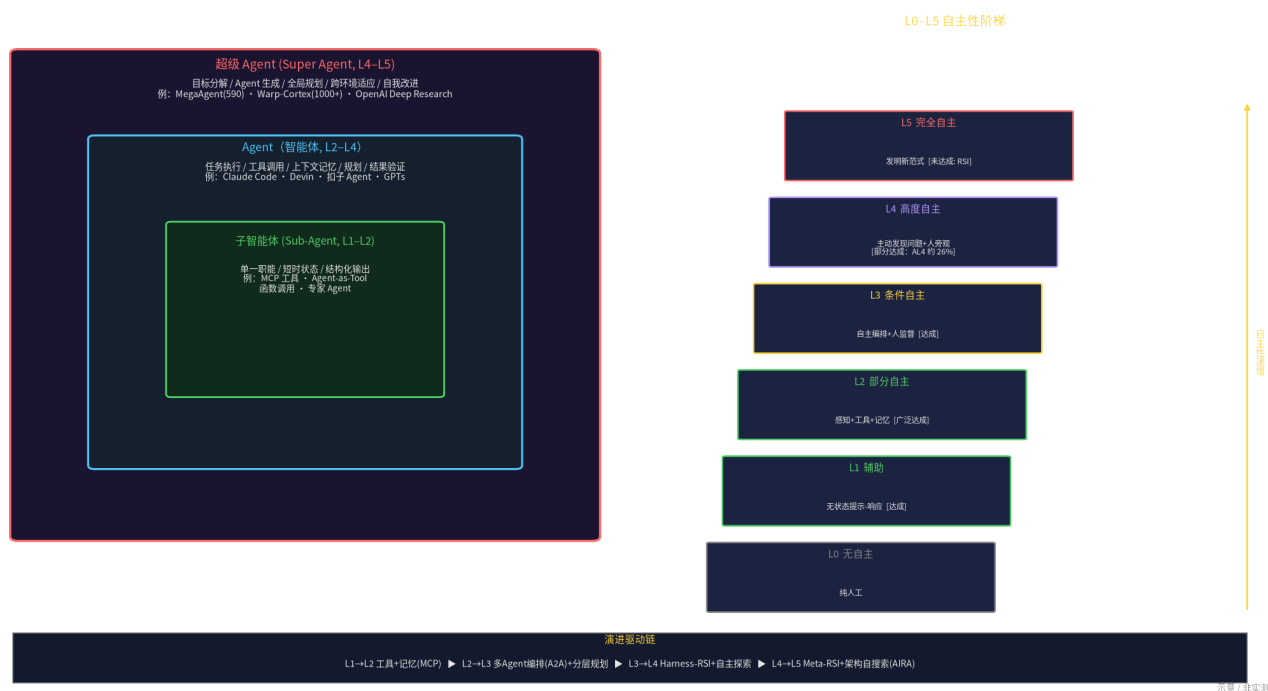
Table 2-1 Five-layer responsibilities × key products × inter-layer interfaces (design proposal • to be implemented)

Layer	Responsibility (what sold)	Key products/ standards (public positioning)	Interface provided to upper layer	Downward dependency
L1 model substrate	Intelligence itself: multimodal, reasoning, graded routing	Doubao/GPT/ Claude/Llama/ Qwen	Unified reasoning API (with routing and cost-capability metadata)	Compute and training
L2 protocol layer	Tool calls / Agent collaboration / security governance	MCP (stateless core, 2026-07-28), A2A v1.0 (Signed Agent Card), SAEP, AGTP; governance AAIF	Standardized tool schema, Agent Card, audit interface	L1 reasoning capability
L3 development and enterprise platforms	Workbench and governance for building Agents	Coze / TRAE / AgentKit / HiAgent 3.0 / ArkClaw / AI Trust / Volcano Ark	Drag/IDE/SDK producing deployable Agents	L1+L2
L4 product layer	Consolidating Agents into role products	Work Agent / companion Agent / phone assistant Agent	Scenario-facing finished Agents	L3
L5 C-end forms	User reach entries	App/mini program, PC/Web, IM/social, browser plugins, smart speakers	Conversation→delegation→autonomy interaction surface	L4

Chapter summary: the five-layer stack is a "layered decoupling" cognitive map—lower layer changes upper layer not rewritten (change model without touching protocol), protocols unified then platforms interchangeable. Note this is cognitive scaffolding, not physically strongly bound five layers; many products cross layers.

3 Deliverable Two: Concept Relationship Chart (Nested Containment and L0–L5 Autonomy Ladder)

First the nesting and ladder.



3.1 Three-Layer Nested Containment

"Sub-agent $\subset$ Agent $\subset$ super Agent (legion)" is a **size nesting** relationship, not three parallel species:

- **Sub-agent (L1–L2):** the narrowest-capability execution unit, called via MCP tools / Agent-as-Tool / function calls / expert Agent forms, finishing work then returning the result.
- **Agent (L2–L4):** an autonomous unit that can independently undertake a segment of tasks, e.g. Claude Code, Devin, Coze Agent, GPTs.
- **Super Agent (L4–L5):** an orchestrator that can autonomously plan and schedule a large group of sub-Agents/Agents, e.g. MegaAgent (590 Agents, ☒ verified), Warp-Cortex 1000+ (**single-author preprint self-report, not independently reviewed**), OpenAI Deep Research.

3.2 L0–L5 Autonomy Ladder (modeled on SAE J3016; corresponding to Epoch AI AL0–AL5, adopted by Anthropic, verified)

Level	Name	Criterion	Status	Representatives
L0	No autonomy	Pure manual	—	Manual process
L1	Assisted	Stateless prompt-response	 achieved	ChatGPT / Doubao
L2	Partial autonomy	Perception + tools + memory	 broadly achieved	Tool-equipped Copilot
L3	Conditional autonomy	Autonomous orchestration + human supervision	 achieved	Software engineering Agent
L4	High autonomy	Proactively finds problems + human observes	 partially achieved	Anthropic discloses about 26% R&D "led" by Claude (AL4), >90% at least AL3, about 30,000 internal Agents concurrent (company disclosure, not independently reviewed)
L5	Full autonomy	Invents new paradigms	 not achieved	RSI research goal

Evolution driver chain: L1→L2 via **tools+memory (MCP)**; L2→L3 via **multi-Agent orchestration (A2A) + layered planning**; L3→L4 via **Harness-RSI + autonomous exploration**; L4→L5 via **Meta-RSI + architecture self-search (AIRA)**. Today the whole industry's most honest coordinate: L3 achieved, L4 just touched the edge (and numbers all self-assessed), L5 still a research goal.

3.3 Two Or Primitives: Agent-as-Tool vs Handoff

On the question "how do multiple Agents collaborate," the industry has converged on two **simultaneously standardized** primitives (Microsoft Agent Framework / AutoGen and OpenAI Agents SDK,  existence verified):

- **Agent-as-Tool:** function call / RPC style. Main Agent retains control, sub-Agent called like a function, after execution **returns result string**, main Agent decides next step.
- **Handoff:** responsibility **formally transferred**—like transferring a phone call/relay race, conversation ownership handed to receiver, **receiver directly faces the user**.

Selection criterion in one sentence: **"I coordinate, you help me compute" uses Agent-as-Tool; "this matter is now yours to manage" uses Handoff**. The former does not release control, the latter transfers control whole-organization, mismatch directly leads to "who on earth is responsible for the result" unclaimed.

Chapter summary: first see clearly the size nesting of "sub/Agent/super," then step firmly on the "L0–L5" autonomy ladder, finally when orchestrating think clearly "whether control stays." Understanding these three things, multi-Agent design won't deviate from the start.

4 Deliverable Three: Six Forms × Nine Dimensions Matrix

First the matrix full picture.

	定位	用户	自主性	执行环境	记忆	协作	权限	场景	入口	
③→④ SOP驱动→目标驱动 ②→③ A2A 演化 ⑤→⑥ 内存架构 Warp-Cortex 演化	① Chatbot	对话问答	C端大众	L1	云端App	会话级	无	只读	客服FAQ	独立App/网页
	② Copilot	嵌入式辅助	知识工作者	L1-L2	IDE/浏览器/Office	项目级	人机协同	读+建议	代码补全/写作	IDE插件
	③ 流程 Agent	固定SOP自动化	运营/业务	L2	RPA/工作流	流程状态	人编排/Agent执行	限定API	审批/报表	企业IM
	④ 自主 Agent	目标驱动自主执行	专业开发者	L3-L4	容器/沙箱	长期记忆+向量DB	自主调工具	工具+API+代码	软件工程/研究	CLI/API/SDK
	⑤ 多 Agent 协作	角色分工团队	工程团队	L3-L4	分布式集群	共享黑板+外部状态	A2A+Handoff/AsTool	角色权限+配额	复杂项目	编排平台
	⑥ 超级 Agent 军团	动态编队大规模自治	企业级/研究级	L4-L5	分布式+边缘+消费硬件	跨应用真可移植记忆	MCP+A2A+SAEP	零信任+RBAC+审计	政策/世界模拟	平台+API+泛在设备

示意 / 非实测

Table 4-1 Six forms × nine dimensions genealogy matrix (design proposal • to be implemented)

Dimension	①Chatbot	②Copilot	③Process Agent	④Autonomous Agent	⑤Multi-Agent collaboration	⑥Super Agent legion
Positioning	Conversational Q&A assistant	Embedded workflow assist	Fixed SOP automation	Goal-driven autonomous execution	Role division team collaboration	Dynamic formation large scale autonomy
Target users	C-end public	Knowledge workers	Operations/business people	Professional developers	Engineering teams	Enterprise/research level
Autonomy level	L1	L1-L2	L2	L3-L4	L3-L4	L4-L5
Execution environment	Cloud/App	IDE/browser/Office	RPA/workflow engine	Container/sandbox	Distributed clusters	Distributed+edge+cloud hardware
Memory	Session level	Project level	Process state	Long-term memory+vector DB	Shared blackboard+external state	Cross-vendor portable memory
Collaboration mode	None	Human-machine collaboration	Human orchestrates/Agent executes	Agent autonomously calls tools	A2A protocol+Handoff/AsTool	MCP+A2A+SAEP three protocol stack
Permissions	Read-only conversation	Read+suggest	Limited API scope	Tools+API+code execution	Role permissions+resource quotas	Zero trust+RBAC+audit chain
Typical scenarios	Customer service/FAQ	Code completion/writing	Approvals/reports	Software engineering/research	Complex project management	Policy simulation/workflow simulation
Entry	Independent App/webpage	IDE plugin/browser extension	Enterprise IM/workflow interface	CLI/API/SDK	Orchestration platform/team kanban	Platform+API+devices ubiquitous

4.1 Three Key Leaps

- **③→④: SOP driven → goal driven.** This is the **core leap**—the process Agent "executes a preset SOP," the autonomous Agent "given a goal, itself decomposes the task." The difference not in model size, but in "who decides next step": SOP is human-written fixed, goal is Agent itself decomposed.
- **④→⑤: catalyzed by the A2A protocol.** A single autonomous Agent can only itself call tools; with A2A this standardized inter-Agent conversation language, multiple Agents can delegate to and discover each other like a team, the multi-Agent collaboration form holds (A2A v1.0 verified).
- **⑤→⑥: catalyzed by memory architecture breakthrough.** To pull collaboration scale from dozens to hundreds-thousands, the bottleneck often not coordination algorithm but memory. Warp-Cortex self-reports pulling single-card concurrency from about 10 to 100+, single-card RTX 4090 only 2.2GB VRAM, theoretical capacity >1000 (**this number single-author preprint self-report, not independently reviewed; and its "98% context compression" not in abstract, this paper does not cite that ratio**). That is, ⑥ is not "pile a few more Agents," but the new form opened only after the memory wall is crossed.

Chapter summary: six forms are a **genealogy**, not six mutually exclusive drawers—a product can grow from ① to ④. When selecting first ask "what autonomy L level, cross-Agent collaboration needed, how large scale," three questions settled, the matrix cells naturally framed.

5 Deliverable Four: Agent Production SOP Swimlane

First the swimlane full picture.



The SOP is **8 stages × 4 roles (PM / systems architect / engineer / QA red team)**, per stage with responsibilities, and gates at key points.

- Requirement input and clarification:** PM produces requirement doc/success criteria/prohibited action list; systems architect does feasibility and model selection judgment. **! human-review gate ① requirement review.**
- Role design and capability definition:** PM defines role responsibilities/user journeys; architect defines Agent topology (star/layered/hybrid) and MCP/A2A selection; engineer produces `role.md` / `agent_spec.yaml` / model routing / tool set.
- Configuration generation and integration:** engineer implements core logic + MCP tools + A2A communication + memory system; architect defines layered topology + circuit breaker thresholds/retry budget/BFT-lite committee; PM aligns TransferBundle six fields (Goal/Context/Done/ToDo/Trace/Owner). **! human-review gate ② architecture review.**
- Permissions and security configuration:** architect configures RBAC + tool whitelist + zero-trust identity; red team designs prompt injection/over-privilege access/data leakage/Byzantine simulation cases. **! human-review gate ③ permission tests.**
- Quality acceptance • mechanically decidable:** red team runs contract tests (Agent-as-Tool five error types / TransferBundle six-field validation / audit invariants) + end-to-end completion tests + fault injection (crash / drop_context / byzantine); engineer fixes, regression all green. **! human-**

review gate ④ acceptance review. This stage acceptance "mechanically decidable"—not relying on subjective scoring, relying on script assertions.

6. **Red-team testing and adversarial verification:** red team attacks semantic common-cause faults/deliberation attacks/Sybil injection/context poisoning; architect does hardening + BFT parameters + common-cause cutting. 🚫 **red-team gate.**

7. **Registration archiving and launch:** engineer registers platform catalog + monitoring alerts + evidence grade marking; PM builds ledger + records known limitations + marks unachieved gates. ⚠️ **human-review gate** ⑤ **launch approval.**

8. **Continuous optimization and version management:** engineer reports run metrics/performance decay/optimization PRs; QA does regression + baseline JSON comparison + version number assertions; architect does structural optimization/topology upgrade. 🔒 **version gate.**

Table 5-1 Gate entry/exit conditions overview (design proposal • to be implemented)

Gate	Position	Type	Entry / exit conditions
① Requirement review	stage1→2	⚠️ human review	Entry: goal mechanically decidable, prohibited action list complete
② Architecture review	stage3→4	⚠️ human review	Exit: fan-in budget/round budget/fault recovery strategy in place
③ Permission tests	stage4→5	⚠️ human review	Exit: Agent cannot execute any operation in "prohibited action list"
④ Acceptance review	stage5→6	⚠️ human review	Exit: completion rate meets target, governance six all zero, Trace complete
🚫 Red-team gate	stage6→7	red team	Exit: no high-risk vulnerabilities, common-cause faults cuttable, Byzantine members cannot force-stop, ledger conservation not broken
⑤ Launch approval	stage7	⚠️ human review	Exit: evidence grades correct, limitations declared, monitoring in place
🔒 Version gate	stage8 continuous	version gate	Baseline metrics cannot silently regress, regression means gate failure, version number automatically asserted

Chapter summary: the whole SOP is **5 human-review gates + 1 red-team gate + 1 version gate**. Its design philosophy "clamp human review at mechanically decidable places"—stage 5 acceptance not relying on subjective scoring, this is the premise for it to scale and be reproducible. Gates not the more the better, too dense kills iteration speed (see Section 7 failure boundaries).

6 Deliverable Five: Ten-Class Component Field Specification Tables

The following are **UDOS design specifications • to be implemented**; example models (GPT-5.3/Codex, Claude, Kimi K3, Qwen3.8-Max, Doubao, DeepSeek-R1, Llama etc.) are **snapshots**, per platform actual model catalog; unverified capability/context numbers marked [CITATION NEEDED], not fabricating benchmarks.

Table 6-1 Agent Configuration (15 fields)

Field name	Description	Required	Example	Configuration platform
agent_id	Unique identifier	yes	spec-kinematics-0	all
name	Display name	yes	Kinematics expert	all
role	Role type	yes	specialist	all
domain	Domain	yes	kinematics	all
model_provider	Model provider	yes	volcengine	all
model_id	Model identifier	yes	doubao-pro-32k	all
system_prompt	System prompt	yes	You are a kinematics expert...	all
temperature	Generation temperature	no	0.3	all
max_tokens	Max output tokens	no	4096	all
tools	Available tool list	no	[mcp://calculator, mcp://solver]	AgentKit / TRAE
memory_config	Memory configuration	no	{type:vector, size:1024}	HiAgent / AgentKit
autonomy_level	Autonomy level	yes	L3	HiAgent / AgentKit
guardrails	Safety guardrails	no	{intent_guard:true, tool_approval:true}	AI Trust
handoff_targets	Handoff targets	no	[qa-committee, orchestrator]	AgentKit / TRAE
evidence_grade	Evidence grade	yes	verified	all

Table 6-2 MCP Tool Configuration (8 fields)

Field name	Description	Required	Example	Configuration platform
tool_id	Tool unique identifier	yes	mcp://database/query	all
name	Tool name	yes	SQL query tool	all
description	Function description	yes	Execute SQL query and return results	all
input_schema	Input parameter Schema	yes	{sql:string, timeout:int}	all
output_schema	Output Schema	yes	{rows:array, columns:array}	all
auth_type	Authentication type	yes	oauth2	AgentKit / AI Trust
rate_limit	Rate limit config	no	100/min	AgentKit
sandbox	Sandbox config	no	{network:false, fs:readonly}	AI Trust

Table 6-3 A2A Communication (6 fields)

Field name	Description	Required	Example	Configuration platform
agent_card	Agent capability card	yes	{name, skills, endpoints}	all
endpoint	Communication endpoint	yes	https://api.example.com/a2a/	all
auth_method	Authentication method	yes	mtls	AgentKit / AI Trust
protocol_version	Protocol version	yes	v1.0	all
skills	Capability list	yes	[task_decompose, code_review]	all
signature	Cryptographic signature	yes	sha256:abc123...	AgentKit / AI Trust

Table 6-4 TransferBundle Handoff (6 fields)

Field name	Description	Required	Example	Configuration platform
goal	Task goal	yes	Solve kinematics equations	all
context	Fact dictionary	yes	{domain:kinematics, units:SI}	all
done	Completed stages	yes	[triage, specialist]	all
todo	To-do stages	yes	[qa_acceptance]	all
trace	Event sequence numbers	yes	[1,5,12,18]	all
owner	Current responsible party	yes	spec-kinematics-0	all

Table 6-5 Circuit Breaker Configuration (7 fields)

Field name	Description	Required	Example	Configuration platform
breaker_id	Circuit breaker identifier	yes	breaker-kinematics-0	AgentKit / HiAgent
min_requests	Minimum requests	yes	2	AgentKit / HiAgent
threshold	Error rate threshold	yes	0.5	AgentKit / HiAgent
cooldown_ticks	Cooldown period	yes	3	AgentKit / HiAgent
half_open_probes	Half-open probe count	no	1	AgentKit / HiAgent
group_isolation_ratio	Group isolation ratio	no	0.5	AgentKit / HiAgent
kill_switch_enabled	Global stop switch	yes	true	AI Trust

Table 6-6 BFT-lite Committee (6 fields)

Field name	Description	Required	Example	Configuration platform
committee_id	Committee identifier	yes	qa-committee-v1	HiAgent
n_voters	Member count	yes	7	HiAgent
f_byzantine	Tolerated Byzantine count	yes	2	HiAgent
quorum_size	Quorum size	yes	5	HiAgent
timeout_ticks	Timeout period	yes	10	HiAgent
equivocation_handling	Equivocation handling	yes	void_round	HiAgent

Table 6-7 Internal Market Settlement (5 fields)

Field name	Description	Required	Example	Configuration platform
budget_total	Total budget	yes	240	HiAgent
reward_per_order	Reward per order	yes	10	HiAgent
slash_rate	Slash rate	yes	1.0	HiAgent
duplicate_policy	Duplicate labor policy	yes	deny_payment	HiAgent
conservation_check	Conservation check	yes	true	HiAgent

Table 6-8 Governance Audit (5 fields)

Field name	Description	Required	Example	Configuration platform
audit_id	Audit identifier	yes	audit-run-001	AI Trust
trace_ledger	Hash-chain Trace	yes	sha256_chain	AI Trust
failure_detectors	Failure detectors	yes	[state_loss, dup_work, no_closer]	AI Trust
owner_registry	Owner registry	yes	{order: stage→agent}	AI Trust
stop_condition	Stop condition	yes	{max_hops:20, done_predicate}	AI Trust

Table 6-9 Evidence Grading (5 fields)

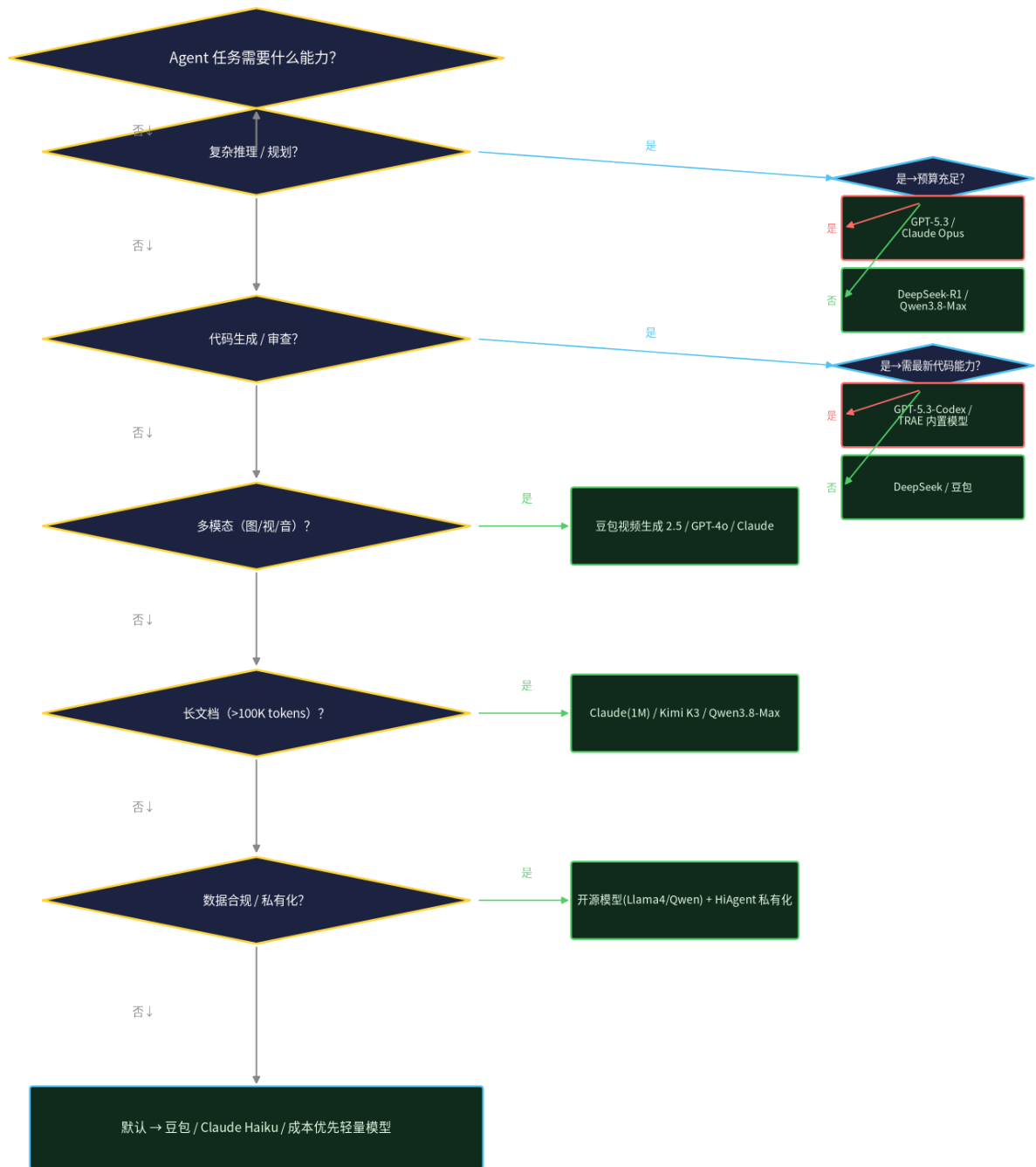
Field name	Description	Required	Example	Configuration platform
evidence_grade	Evidence grade	yes	cpu-proto	all
seed_count	Seed count	yes	1	all
reproducible	Reproducible	yes	true	all
source_json	Source JSON path	yes	reports7/matrix.json	all
ci_gate	CI gate	no	true	HiAgent / AI Trust

Table 6-10 Model Selection Decision Table (7 rows)

Decision dimension	Recommended model	Applicable scenarios	Configuration platform
Complex reasoning / planning	GPT-5.3 / Claude Opus [self-report snapshot]	Task decomposition, strategy design	all
Code generation / review	GPT-5.3-Codex / DeepSeek-R1 [self-report snapshot]	Software engineering tasks	TRAE
Fast response / simple tasks	Doubao / Claude Haiku	Classification, extraction, formatting	Coze / AgentKit
Multimodal understanding	Doubao video generation 2.5 / GPT-4o	Image/video analysis	AgentKit
Long context	Claude (1M context) / Kimi K3 [CITATION NEEDED]	Long document analysis	HiAgent
Cost sensitive	DeepSeek / Qwen	Large-scale deployment	all
Privatized deployment	Open-source models (Llama / Qwen)	Data compliance requirements	HiAgent

Note: GPT-5.3-Codex additionally has OpenAI 2026-02-05 official statement "our first model that was **instrumental in creating itself**" (company disclosure, not independently reviewed), this paper only registers it as an early RSI signal, not expanding into selection basis.

Finally landing on one selection decision tree.



示意 / 非实测；示例型号为快照，以平台实际模型目录为准

Figure 5 Model selection decision tree: first branch by task difficulty/modality/context length/cost/compliance, landing on specific model families and configuration platforms. Concept schematic, not measured; model catalog updates with vendors.

Chapter summary: ten-class field sets consolidate all preceding abstractions (Agent, tools, collaboration, handoff, fault tolerance, consensus, settlement, audit, evidence, selection) into "cells fillable on configuration platforms." Fields uniformly **design proposals • to be implemented**; model columns are snapshots, unverified numbers like long context hang [CITATION NEEDED], absolutely not fabricating benchmarks.

7 Dross, Failure Boundaries and Falsifiability Conditions

7.1 This Paper's Own Dross and Boundaries

- **Five-layer layering is cognitive scaffolding, not strong binding.** Real products often cross layers (one platform both produces Agents and directly reaches C-end); treating five layers as "physical layers that must be strictly isolated," draws charts detached from engineering reality.
- **Six forms are a genealogy, not mutually exclusive categories.** One system can smoothly grow from Chatbot to autonomous Agent; using it for an "either-or" classification table is misuse.
- **SOP gates too dense kill iteration speed.** 5 human-review gates + 1 red team + 1 version gate suits high-risk launch, but may make every change of an internal small tool run the full process; gate density should scale with risk level, not one-size-fits-all.
- **All new architectures/fields are design proposals • to be implemented.** Reading the "specification tables" as "already launched and measured," this paper's biggest misuse risk.
- **Industry numbers all carry discounts.** Warp-Cortex's 100 concurrency/2.2GB/>1000 is single-author preprint self-report, not independently reviewed (its "98% context compression" not in abstract, this paper not adopted); Anthropic's 26%/>90%/about 30,000 Agents is company disclosure, not independently reviewed; IDC's HiAgent 17.8% privatization / Coze 19.3% public cloud is report via media retelling, original not pulled. The above all do not constitute market endorsement.

7.2 Falsifiability Conditions

This paper's architecture claims should be overturned or corrected if the following occur:

- **F1 (layering fails):** if future mainstream products long eat the whole market in "single-layer end-to-end" form, inter-layer protocols no one cross-layer reuses, then the five-layer stack degenerates to post-hoc description, should no longer serve as construction blueprint.
- **F2 (genealogy collapses):** if some single product form is simultaneously optimal on all nine dimensions, smoothing the Chatbot-to-legion gap, then the six-form genealogy loses distinguishing meaning.
- **F3 (gate back-action):** if measurements show gates drag delivery cycle to negative return, and cannot significantly reduce online incident rate, then the SOP should change to "sample review by risk level" rather than full gates.
- **F4 (protocols not complementary):** if MCP/A2A/SAEP three in actual landing are swallowed by some single protocol, then the "three protocols complementary" judgment needs rewriting.
- **F5 (L4 misjudgment):** if Anthropic's AL4 "led" caliber is substantially revised down by external independent review, then this paper's L4 "⚠️ partially achieved" optimism needs lowering.

8 Conclusion

Production-grade multi-agent competition has shifted from "whose model is smarter" to "whose stack is layered more clearly, forms seen more accurately, production gates harder, components configured more reproducibly." This paper takes this apart into five blocks: one **five-layer vertically integrated stack** (model substrate/protocols/platforms/products/C-end), one **L0-L5 autonomy ladder and sub-Agent<super nesting**, one **six forms × nine dimensions genealogy**, one **8-stage + 7-gate**

production SOP, and one **ten-class component field specification**. MCP/A2A/SAEP three protocols complementary, Handoff and Agent-as-Tool two orchestration primitives, ⑤→⑥ relying on memory architecture rather than piling headcount—these are this paper's three engineering judgments most wanted to be remembered.

This is a construction checklist with falsifiability conditions, not a delivered report card; all new architectures to be implemented, all external numbers carry three-state discounts.

References and Evidence Sources

UDOS v7.7.1 supplemental volume (this series)

1. Master layout: P33 "v7.6.0 Architecture Master: Eight Structural Change Specifications and Compatibility Paths" (this volume follows its logo header, structured abstract, tables and dross/falsifiability style).
2. Evidence ledger: `EVIDENCE_LEDGER_v771.md` (verified 2026-09-21, three-state caliber).

Existence verified (✓)

1. Anthropic / A2A / MCP / A2A: MCP official blog "The 2026-07-28 Specification" (stateless core / standard HTTP load balancing); A2A official "A2A Protocol Ships v1.0" (Signed Agent Card); Linux Foundation A2A announcement (established 2025-12) and "A2A Joins A2A" (merged 2026-08-17).
2. Volcano Engine product matrix: AgentKit / HiAgent 3.0 / ArkClaw / TRAE / Coze (volcengine.com public product list, only describes public positioning).
3. Orchestration primitives: OpenAI Agents SDK and Microsoft AutoGen official docs distinction of Handoff vs Agent-as-Tool.
4. MegaAgent: MegaAgent: A Large-Scale Autonomous LLM-based Multi-Agent System Without Predefined SOPs, Findings of ACL 2025 (590-Agent national policy simulation).
5. Autonomy grading: Epoch AI AL0–AL5 automation grading officially adopted by Anthropic.

Paper/company disclosure • not independently reviewed (🟡)

1. Warp-Cortex, arXiv:2601.01298 (single-author preprint; 100 concurrency @ 2.2GB VRAM, theoretical >1000 self-reported; "98% context compression" not in abstract).
 2. Anthropic "Measurements for understanding the pace of AI development inside frontier labs" (2026-09): about 26% R&D AL4 led, >90% at least AL3, about 30,000 internal Agents, company self-assessment, not independently reviewed.
 3. OpenAI "Introducing GPT-5.3-Codex" (2026-02-05): "instrumental in creating itself," company self-statement.
 4. IDC "China Agent Development Platform Market Share, 2025": HiAgent 17.8% privatization / Coze 19.3% public cloud, via media retelling, original not pulled.
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Evidence Discipline and Reproduction Notes

- This paper is an **industry architecture specification and positioning document**, no new experiments; all layering, matrices, SOP gates, ten-class fields are **design proposals • to be implemented**, prohibited from writing as landed measured returns.

- External facts marked per EVIDENCE_LEDGER_v771.md three states:  existence verified /  paper or company disclosure • not independently reviewed / via media retelling • original not pulled; unverified capability numbers hang [CITATION NEEDED], not fabricating benchmarks.
- Industry products only describe public positioning, not endorsing any vendor, not fabricating market share; model models are snapshots, per platform actual model catalog.
- This paper's falsifiability conditions F1–F5 for post-hoc scoring; five-layer stack as cognitive scaffolding, six forms as genealogy not mutually exclusive categories, SOP gate density should scale with risk, all three declared in Section 7.